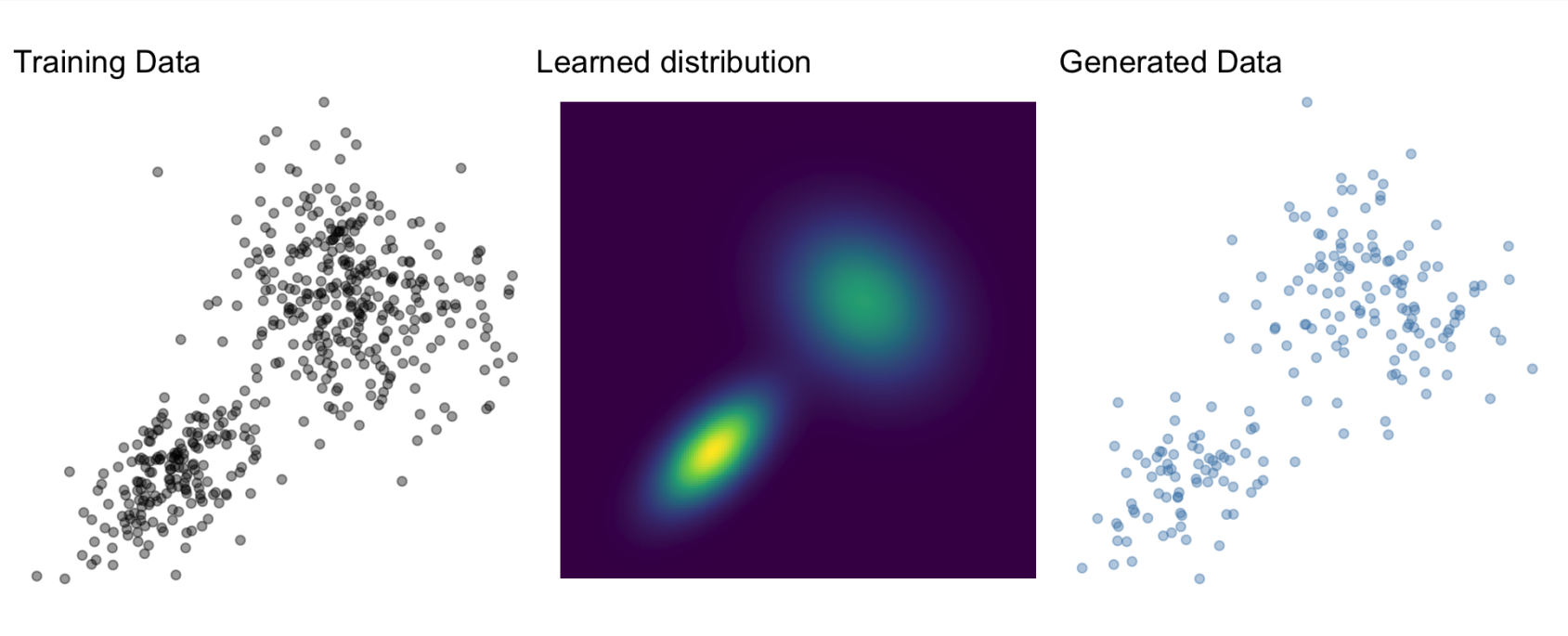
Generative Modeling with Normalizing Flows

Paul Bürkner based on materials of Šimon Kucharský

Generative models

Learn p_X given a set of training data $x_i, \dots, x_n$



- Sampling $x \sim p_X$
- Density evaluation $p_X(x)$

Mixture model

Weighted sum of multiple simpler distributions, e.g., Normal

$$p_X(X) = \sum_k^K w_k \times \text{Normal}(X; \mu_k, \sigma_k)$$

- Sampling and evaluating straightforward
- Theoretically can represent any distribution
- Practically, does not scale well

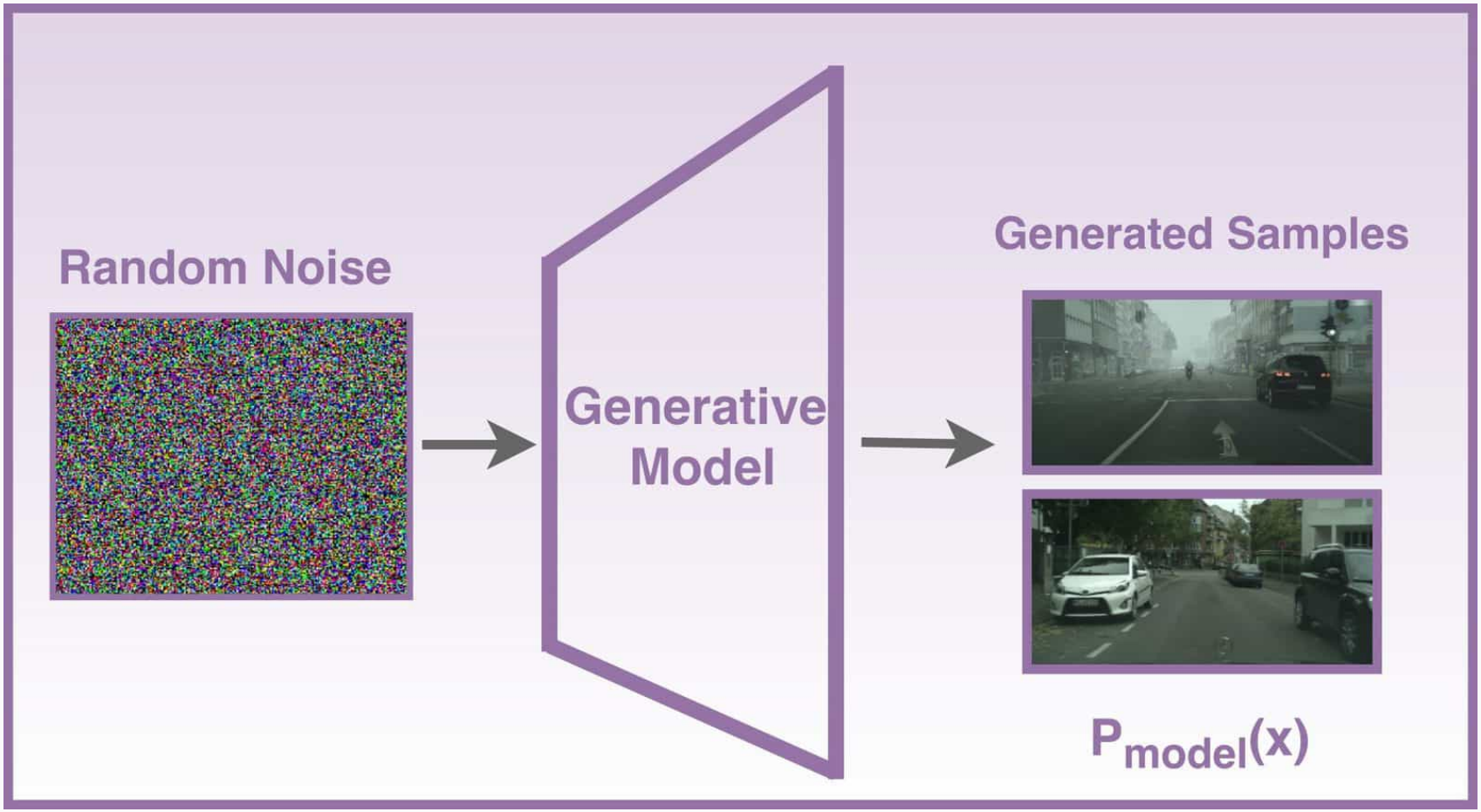
Many architectures

- Markov random fields (Li, 2009)
- Generative adversarial networks (GAN, Goodfellow et al., 2014)
- Variational autoencoders (VAE, Kingma et al., 2013)
- **Normalizing flows** (Kobyzev et al., 2020; Papamakarios et al., 2021)
- Diffusion models (Song et al., 2020)
- Flow matching (Lipman et al., 2022)
- Consistency models (Song et al., 2023)

Common idea

Map p_X to a base distribution p_Z through some operation g

$$x \sim g(z) \text{ where } z \sim p_Z$$



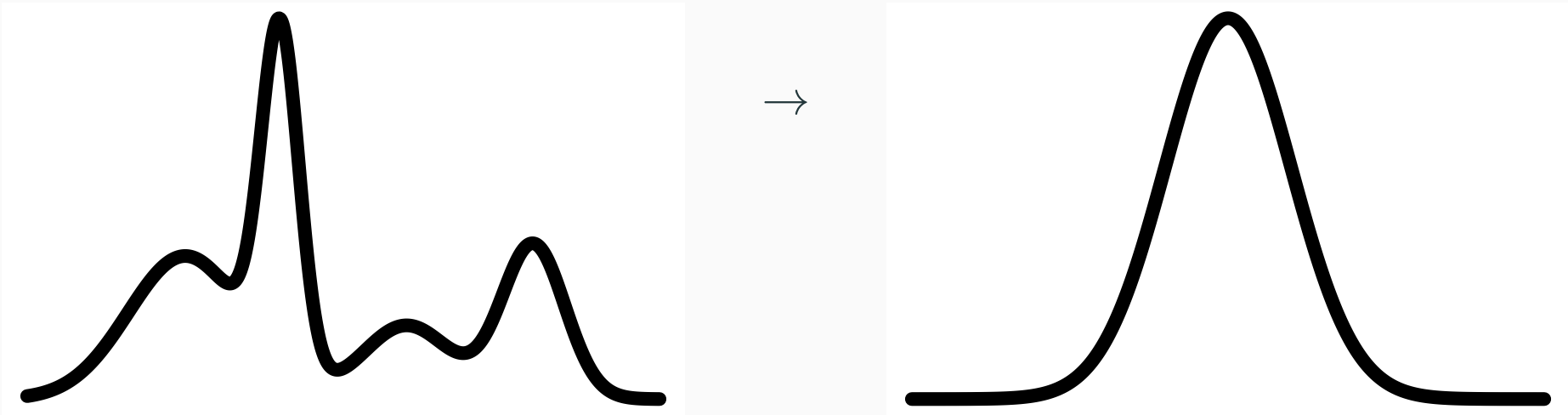
Normalizing flows

Normalizing flows

Built on invertible transformations of random variables

- Find f such that $f(X) = Z \sim \text{Normal}(0, I)$
  - *f normalizes X*

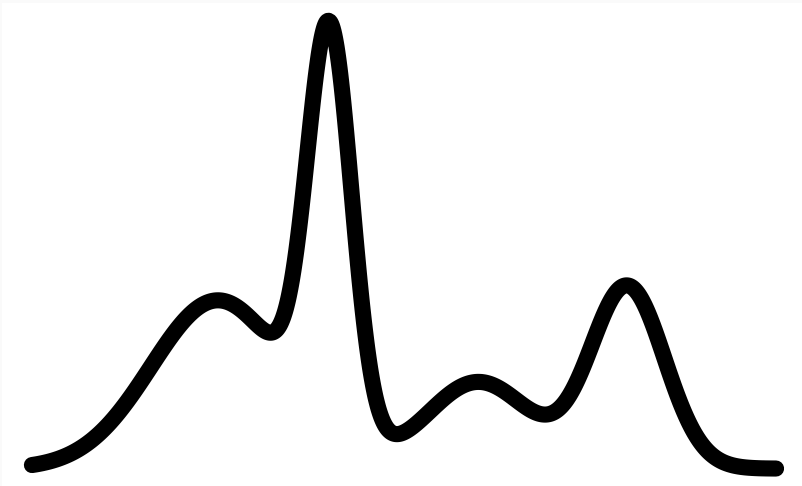
f



Sampling

- Sample $z \sim p_Z$ (e.g., Normal)
- Obtain $x = f^{-1}(z)$

f^{-1}



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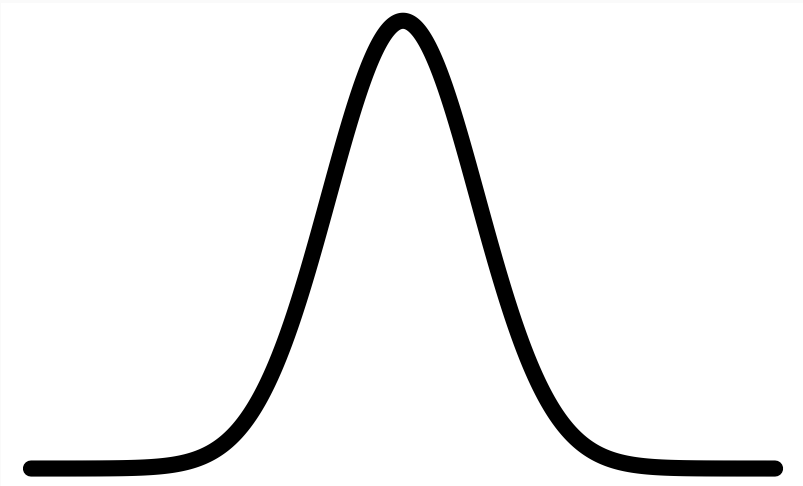

Density evaluation

Change of variables formula

$$p_X(x) = p_Z(f(x)) \left|\det J_f(x)\right|$$

- Express p_X using p_Z and the transform f
- $\left|\det J_f(x)\right|$: Absolute value of the determinant of the Jacobian matrix
  - "Jacobian" for short
  - Volume correction term

Change of variables - intuition

$$Z \sim \text{Uniform}(0, 1)$$

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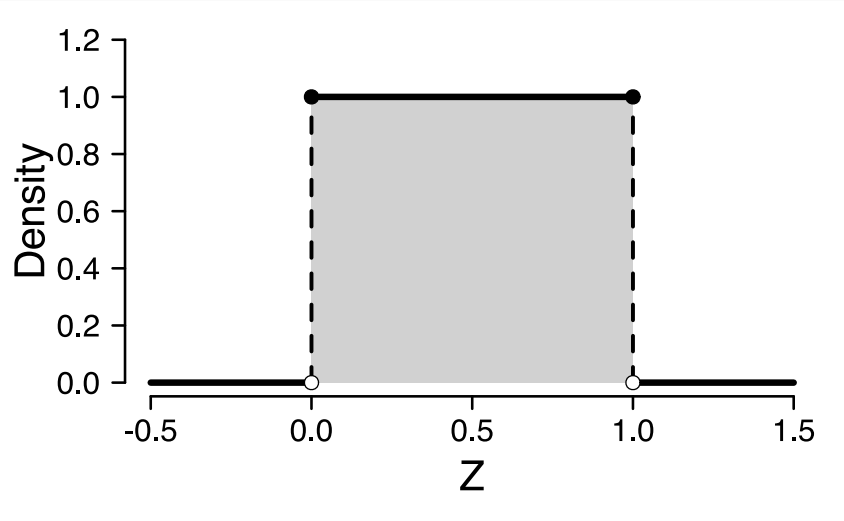

Change of variables - intuition

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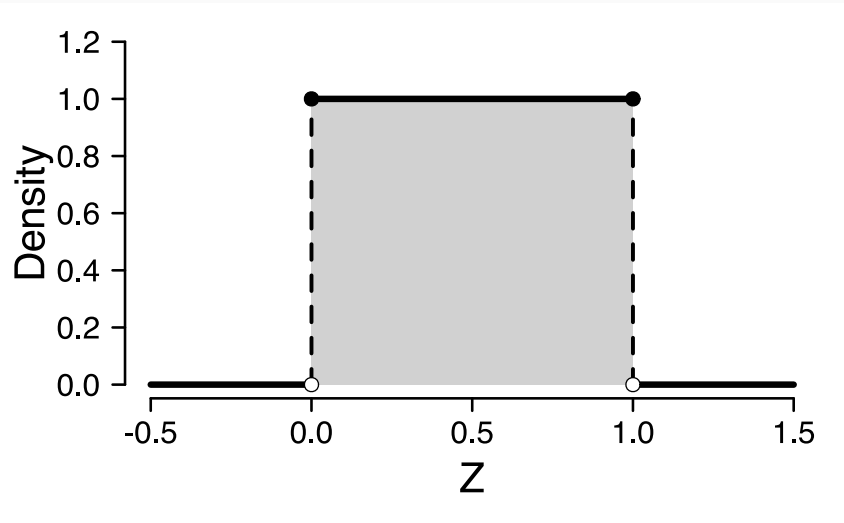
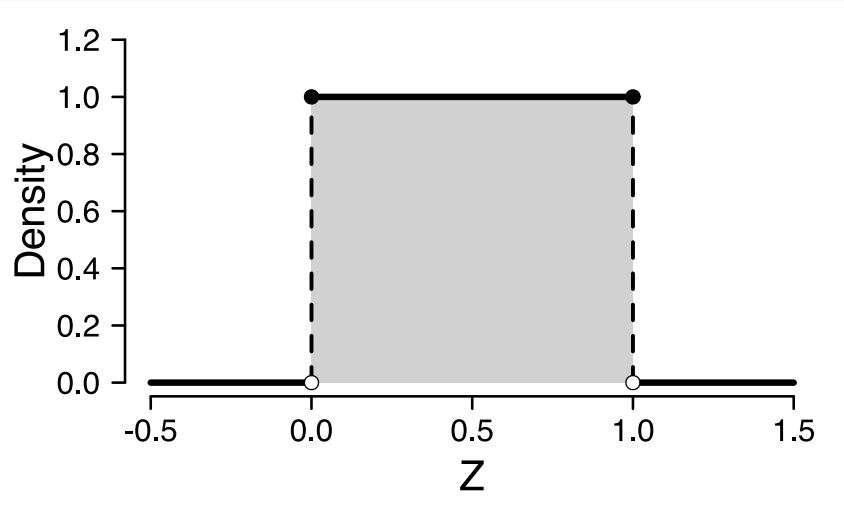


$$X = 2Z - 1$$

Change of variables - intuition

$$Z \sim \text{Uniform}(0, 1)$$



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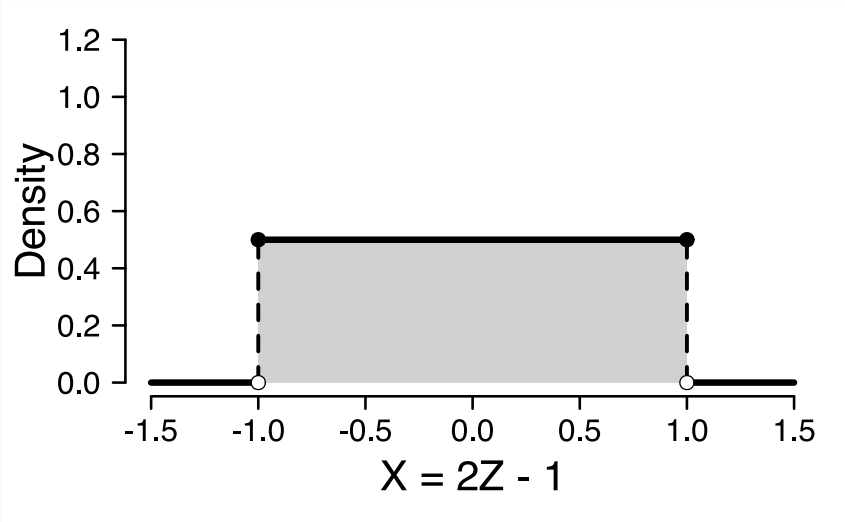


Change of variables - affine transform

$$f : Z = aX + b$$

- shift by b: no effect
- scale by a constant a: multiply by a

$$p_X(x) = p_Z(f(x)) \times a$$

Change of variables - affine transform

Example

$$p_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} z^2\right)$$

$$f : Z = \frac{(X - \mu)}{\sigma}$$

$$\begin{aligned} p_X(x) &= p_Z(f(x)) \times a \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} f(x)^2\right) \times a \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right) \end{aligned}$$

Change of variables - more formally

$$p_X(x) = p_Z(f(x)) \left| \frac{d}{dx} f(x) \right|$$

Change of variables - more formally

$$p_X(x) = p_Z(f(x)) \left| \frac{d}{dx} f(x) \right|$$

Example

$$f : Z = \log(X)$$

$$\frac{d}{dx} f(x) = \frac{d}{dx} \log(x) = \frac{1}{x}$$

$$p_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} z^2\right)$$

$$p_X(x) = \frac{1}{x\sqrt{2\pi}} \exp\left(-\frac{1}{2} \log(x)^2\right)$$

Change of variables - multivariate

$$p_X(x) = p_Z(f(x)) \left|\det J_f(x)\right|$$

$$J_f(x) = \begin{bmatrix} \frac{\partial z_1}{\partial x_1} & \cdots & \frac{\partial z_1}{\partial x_K} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_K}{\partial x_1} & \cdots & \frac{\partial z_K}{\partial x_K} \end{bmatrix}$$

Change of variables - multivariate

$$f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1^2 x_2 \\ 3x_1 + \sin x_2 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$J_f(x) = \begin{bmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 x_2 & x_1^2 \\ 3 & \cos x_2 \end{bmatrix}$$

Normalizing flow

$$p_X(x) = p_Z(f(x)) \left|\det J_f(x)\right|$$

Define a f as a neural network with trainable weights ϕ

Training

Maximum likelihood (or rather: negative log likelihood)

$$\arg\min_{\phi} - \sum_{i=1}^{n} \log p_Z(f(x_i \mid \phi)) + \log \left|\det J_f(x_i \mid \phi)\right|$$

Flow f

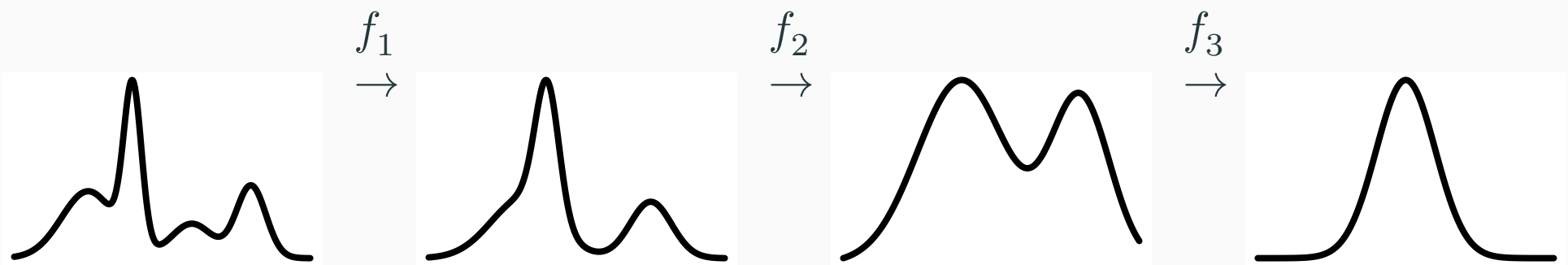
Challenge

- Sampling: Invertible (f^{-1})
- Training:
  - Differentiable
  - Computationally efficient Jacobian
- Expressive to represent non-trivial distributions

Flow composition

Invertible and differentiable functions are “closed” under composition

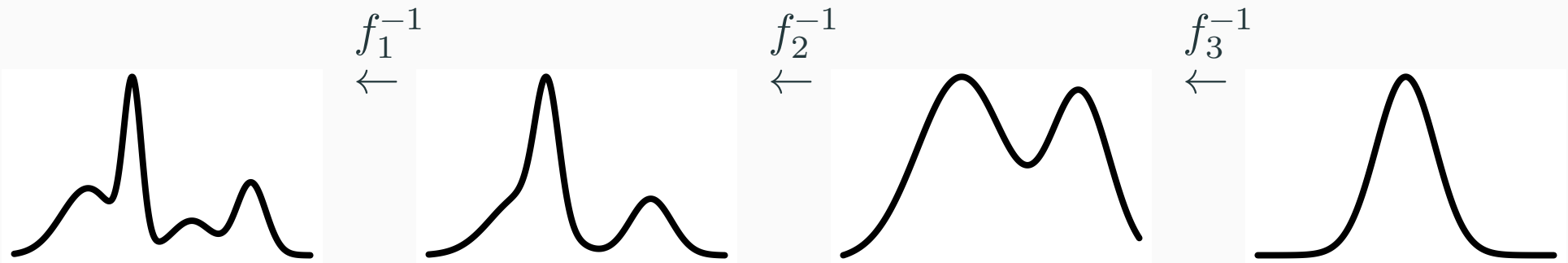
$$f = f_L \circ f_{L-1} \circ \dots \circ f_1$$



Flow composition - inverse

To invert a flow composition, we invert individual flows and run them in the opposite order

$$f^{-1} = f_1^{-1} \circ f_2^{-1} \circ ... \circ f_L^{-1}$$



Flow composition - Jacobian

- Chain rule

$$\left|\det J_f(x)\right| = \left|\det \prod_{l=1}^{L} J_{f_l}(x)\right| = \prod_{l=1}^{L} \left|\det J_{f_l}(x)\right|$$

- if we have a Jacobian for each individual transformation, then we have a Jacobian for their composition

$$\arg\min_{\phi} \sum_{i=1}^{n} \log p_Z(f(x_i \mid \phi)) + \sum_{l=1}^{L} \log \left|\det J_{f_l}(x_i \mid \phi)\right|$$

Linear flow

$$f(x) = Ax + b$$

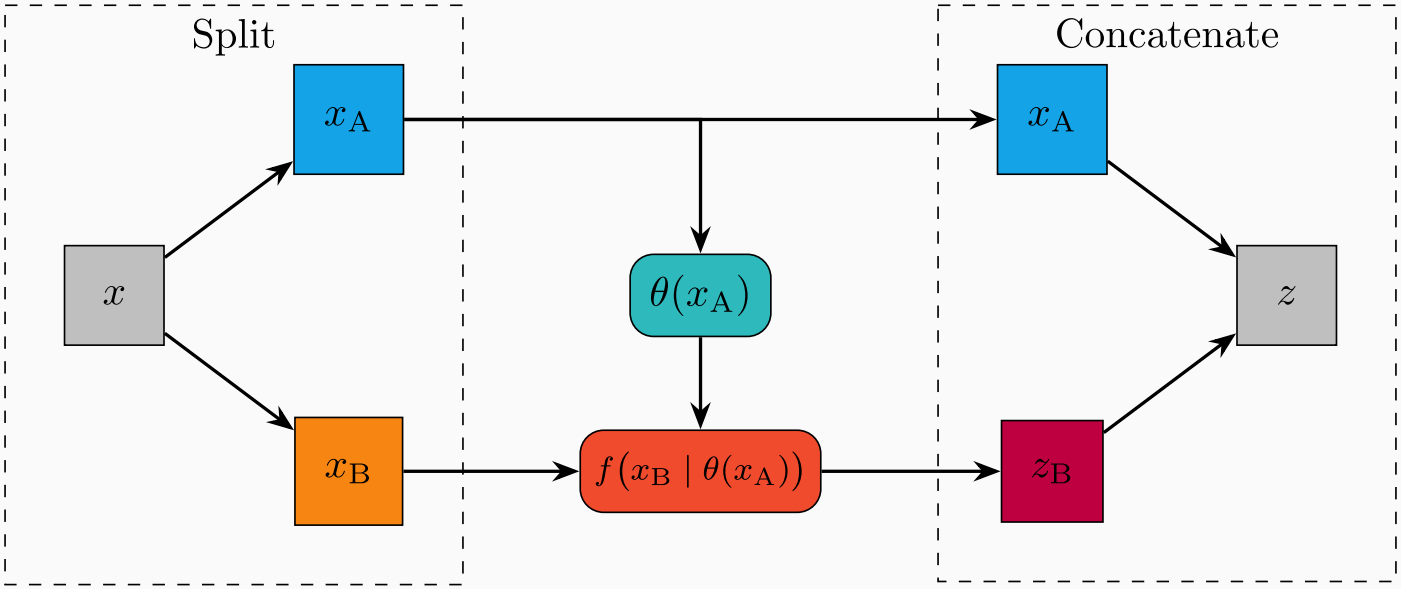
- inverse: $f^{-1}(z) = A^{-1}(x - b)$
- Jacobian: $\left|\det J_f(x)\right| = |\det A|$
- Limitations:
  1. Not expressive (composition of linear functions is a linear function)
  2. Jacobian/inverse may be in $\mathcal{O}(p^3)$

Coupling flows

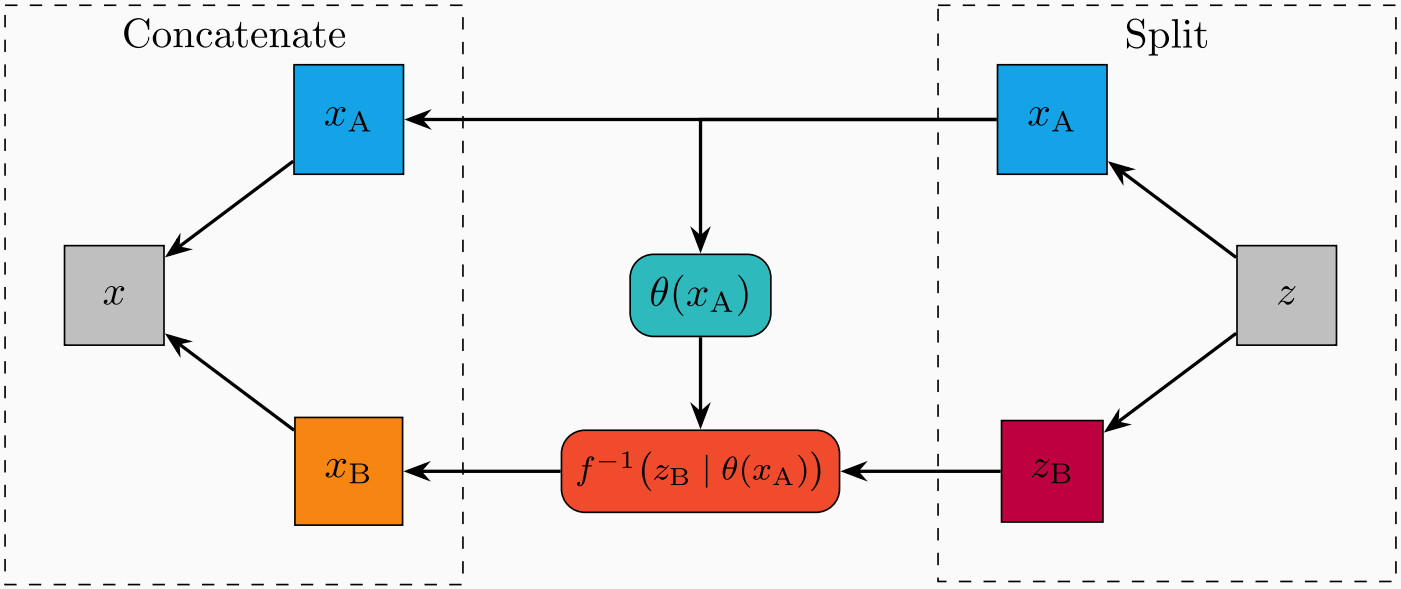
- *Increasing* expresiveness while potentially *decreasing* computational costs
- A *coupling flow* is a way to construct non-linear flows

1. Split the data in two disjoint subsets: $x = (x_A, x_B)$
2. Compute parameters conditionally on one subset: $\theta(x_A)$
3. Apply transformation to the other subset: $z_B = f(x_B \mid \theta(x_A))$
4. Concatenate $z = (x_A, z_B)$

Coupling flow: Forward



Coupling flow: Inverse



Coupling flow trick

- Jacobian

$$J_f = \begin{bmatrix} \mathbb{I} & 0 \\ \frac{\partial}{\partial x_A} f(x_B \mid \theta(x_A)) & J_f(x_B \mid \theta(x_A)) \end{bmatrix}$$

- Determinant

$$\det J_f = \det(\mathbb{I}) \times \det J_f(x_B \mid \theta(x_A)) = \det J_f(x_B \mid \theta(x_A))$$

Coupling flow trick

- $f(x_B \mid \theta(x_A))$ needs to be
  - differentiable
  - invertible
  - easy determinant of the Jacobian
- $\theta(x_A)$ can be arbitrarily complex
  - non-linear
  - non-invertible
  - $\rightarrow$ neural network
- Stack multiple coupling blocks and permute x_A and x_B

Affine coupling (Dinh et al., 2016)

- $\theta(x_A)$: Trainable coupling networks, e.g., MLP
    - Output: Shift μ and scale σ
- Linear (affine) transform function $f(x_B \mid \theta(x_A)) = \frac{x_B - \mu(x_A)}{\sigma(x_A)}$
- Jacobian: $-\log \sigma(x_A)$

Spline coupling (Müller et al., 2019)

- Transformation: Splines
  - "Piecewise polynomials"
- More expressive
- Easier to overfit
- Slower at training and inference

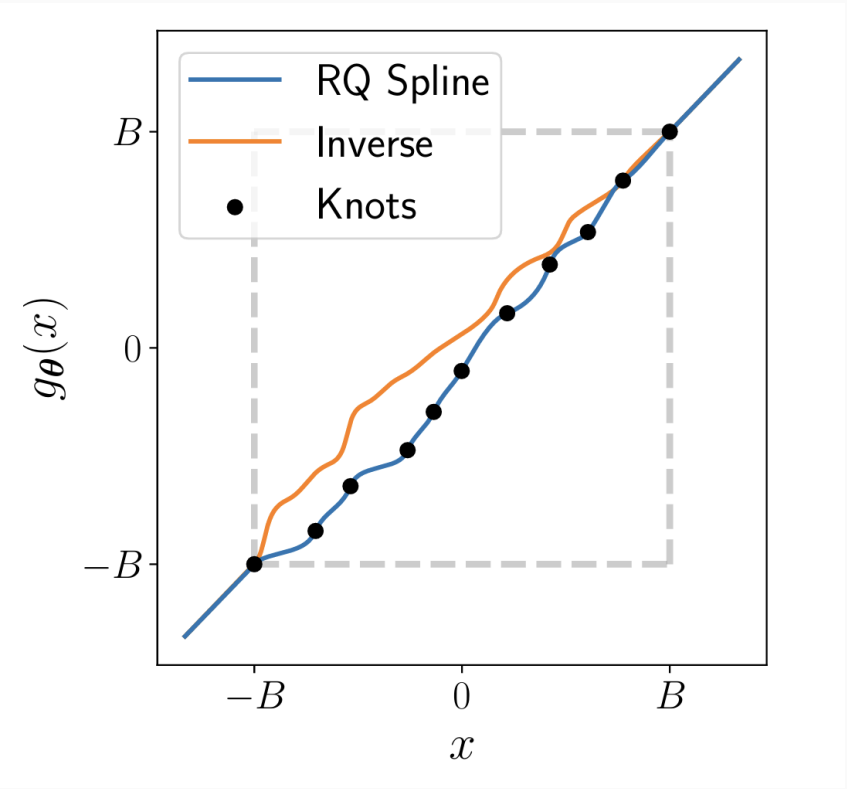


Figure 2: Figure from Durkan et al. (2019)

Build your own normalizing flow

Time for exercise `normalizing_flow`

References

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